

Project: Benchmarking detection of DNA copy number variation using high-density methylation microarrays

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Data source: FinalSet

Central Thesis

The accuracy of methylation-based CNV calling tools is not determined by software choice or parameter tuning, it is determined by the **genomic complexity of the tumour**, as quantified by the Genomic Index (GI).

In low-complexity benign tumours (LM), all three tools perform adequately. In high-complexity malignant tumours (LMS), all three tools fail **proportionally** to the degree of genomic rearrangement.

At-a-glance stats

Metric	Value
LMS mean accuracy (all tools)	53–58%
LM mean accuracy (all tools)	92–96%
GI-accuracy Pearson r	–0.66
GI seen by tools vs SNP	13–22% of true GI

1. Introduction

Structure

- LMS and LM are smooth-muscle tumours with opposite genomic profiles. LMS is malignant and has high genomic complexity; LM is benign and has low genomic complexity.
- SNP arrays are the gold standard for CNV detection but require fresh tissue and specialised infrastructure. Methylation arrays (EPIC) are increasingly used for CNV calling because the data is already being generated for epigenetic profiling. No benchmarking study has systematically compared the two platforms across the full benign-to-malignant spectrum.
- **New Contribution:**

1. Accuracy benchmarking across three tools and four bin sizes
 2. A mechanistic explanation via the Genomic Index
 3. Gene-level concordance for clinically relevant loci
 4. A null result on bin size that rules out a parameter explanation for observed failures
- The accuracy numbers reported are concordant with an [existing paper](#). The novel contribution is not the accuracy numbers themselves but that the scope of use might be limited by the genomic complexity.

Key citations needed

- CBS (circular binary segmentation) algorithm
 - Sesame, MethylMaster, and Conumee tool papers
 - Concordant accuracy paper — introduce here, revisit in Discussion
 - GI as a measure of genomic complexity — define formally here or in Methods
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2. Methods

What to include

- **Dataset:** 13 LMS cases and 9 LM cases. All processed via the EPIC methylation array. SNP data from OncoScan CNV arrays (gold standard).
- **Tools evaluated:** Sesame, MethylMaster, Conumee, each at default + four bin sizes (10 kb, 50 kb default, 100 kb, 1 Mb).
- **Preprocessing:** Raw data converted to standardised GenomicRanges dataframes.
- **Genomic Index:** It penalises both the number and amplitude of copy number events. Formula below.

$$GI = \frac{(\sum \#CNVs)^2}{\# \text{ of chromosomes showing somatic changes}}$$

- **Accuracy metric 1 — whole-dataset:** Overlapping predicted and truth GenomicRanges, matching basepairs divided by total genome assessed. Mean taken across all cases.
 - **Accuracy metric 2 — median:** Same calculation, but median basepair accuracy per case.
 - **Gene concordance:** Seven genes assessed: CDKN2A, CCNE1, MYC, MYOCD, PTEN, RB1, TP53. CNV status from each methylation tool vs SNP ground truth.
 - **Genome modified:** Percentage of genome covered by called CNV segments per case and tool. Assesses detection sensitivity independent of call accuracy.
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3. Results

3.1 Accuracy splits cleanly by tumour type

Numbers to cite

Tool	LMS mean accuracy	LM mean accuracy
Conumee	58.4%	91.6%
MethylMaster	54.8%	96.4%
Sesame	52.8%	95.9%

- **LM median accuracy >99.8%** for all tools. LM failures are concentrated in a small number of outlier cases (primarily Case 9312, LM outlier), not a systematic failure.
- **Tool ranking flips between cohorts.** Conumee leads in LMS but is worst in LM. MethylMaster leads in LM.

Figures

- **Fig. 5 (poster)** — Bin-wise accuracy bar chart, LMS. Point out that cases 9338, 9328, 9350 cluster at the bottom regardless of tool or bin.
- **Fig. 6 (poster)** — Bin-wise accuracy bar chart, LM. Highlight consistency, with 9312 as the visible exception.

3.2 Bin size does not rescue accuracy in LMS

A null result.

Bin size	LMS mean accuracy	Note
10 kb	54.2%	Baseline
50 kb	55.3%	No improvement
100 kb	55.2%	No improvement
1 Mb	56.7%	Marginal — not meaningful

Accuracy is flat across two orders of magnitude. If CBS failure were a resolution problem, you would expect a clear recovery at 1 Mb. The 2.5 pp increase at 1 Mb is not meaningful.

This means that there is no explanation on the basis of the bin parameter.

Figures

- **Fig. 5 & 6 (poster)** — Highlight in text that bars are nearly identical across bins within each case.

3.3 Genomic Index explains the accuracy gradient

Argument built in 3 steps:

1. GI separates cases by complexity
2. Accuracy decreases with GI
3. Methylation tools systematically underestimate GI.

Case-level LMS table (source: `GenomicIndexLMS.csv` + `accuraciesLMS.csv`)

Case	SNP GI	SNP genome %	Mean accuracy	Flag
9350	3,463	40.9%	31.6%	Highest GI in cohort, re-verified
9328	3,388	22.3%	28.1%	Extreme GI, re-verified
9338	3,002	16.8%	24.2%	Worst overall, re-verified
9357	917	37.7%	54.7%	Average
9356	666	41.5%	56.0%	Average
9353	550	47.3%	48.1%	Average
9355	511	42.8%	53.3%	Average
9354	393	45.7%	56.0%	Average
9327	464	42.1%	52.4%	Average
9358	868	44.3%	45.0%	Average
9337	2,200	25.1%	77.7%	ANOMALY
9202	60	16.7%	94.3%	LM-equivalent GI, re-verified by hand
9203	56	21.4%	98.1%	LM-equivalent GI, re-verified by hand

Pearson $r = -0.659$ (SNP GI vs mean accuracy across LMS cases)

Pearson $r = -0.652$ (SNP GI vs mean accuracy across LM cases)

This shows decent correlation.

GI compression ratios (source: `GenomicIndexLMS.csv` + `genomeChangedLMS.csv`)

Tool	GI seen as % of SNP GI	Genome territory detected	Failure mode
MethylMaster	21.6%	96.4% of SNP territory	Finds regions; misses CNV depth
Conumee	16.2%	76.1% of SNP territory	Partial territory + depth loss
Sesame	12.8%	49.2% of SNP territory	Misses half the affected genome

🔗 Two mechanistically distinct failure modes: **MethylMaster** correctly identifies 96% of the genome alteration but compresses copy number amplitude. **Sesame** fails at the location level, detecting only 49% of affected genome territory, and also compresses amplitude. This can be verified with a manual observation of the graphs generated.

Case 9337, anomalous

✍ **SNP GI = 2,200, but achieves 77.7% mean accuracy.**

Methylation tools detect *more* genome territory than SNP does in this case (MethylMaster ratio = 1.24×, Conumee = 1.26×). Manual observation showed that the high GI was actually driven by a sort of extreme segmentation in the SNP data, which means that the value was extremely inflated, **HOWEVER**, notably, they were in widely segmented, so by just measuring deletions and amplifications, it gives a higher accuracy.

Implication: GI is a necessary but not sufficient predictor of tool performance.

Cases 9202 and 9203

SNP GIs of 60 and 56, close to the LM cohort (median SNP GI ~8). Accuracies of 94.3% and 98.1% track accordingly.

Including them in LMS summary stats artificially elevates the mean.

Options to mitigate statistical inaccuracy:

- Report LMS accuracy with and without these two cases
- Flag them as a biologically distinct sub-group of low-complexity LMS

Figures

- **Fig. 7 (poster)** — LMS and LM metrics heatmap.

3.4 Gene concordance reveals selective blind spots

Gene concordance table

Gene	Concordance	SNP ground truth	Interpretation
CDKN2A	87% ✓	Deleted 12/13 LMS	Large canonical deletion
TP53	85% ✓	Amplified 9/13 LMS	Strong amplification signal
PTEN	74%	Mixed	Acceptable
RB1	62%	Mixed	Borderline
MYC	41% ✗	Focal amplification	Near-random
CCNE1	41% ✗	Focal amplification	Near-random
MYOCD	36% ✗	Focal	Most significant miss

- **Deletion vs amplification:** CDKN2A (deletion, 87%) and TP53 (amplification, 85%) are captured with similar success. This means that the distinction between the difficulty of calling amplification v. deletion might not necessarily be significantly different.

Figures

- Fig. 4 (poster) — Gene concordance heatmap.

4. Discussion

Outline

¶1 — **Central finding:** Methylation-based CNV calling in LMS performs near chance (~53–58%), while LM performance is clinically adequate (~92–96%).

¶2 — **Connect to the concordant prior paper:** Your accuracy numbers reproduce those reported by [Smith et al., 2021](#).

¶3 — **The CBS failure mechanism:** CBS segments by identifying statistically significant transitions in log2 copy number ratios. In a genome where 25–47% of chromosomal territory is altered (mid-range LMS cases), the noise floor is elevated globally. The null result on bin size (flat from 10 kb to 1 Mb) confirms this is not a resolution artefact.

¶4 — **Two failure modes of tools:** MethylMaster and Sesame represent mechanistically distinct failure modes:

- MethylMaster: high sensitivity to the genome modified (96% territory), compresses amplitude

- Sesame: misses 51% of genome modified, also compresses amplitude

Future improvement efforts should target different aspects of each tool's pipeline.

¶5 — The 9337 anomaly and CNV architecture: Case 9337 demonstrates that aggregate GI is not the only predictor. Despite SNP GI of 2,200, it achieves 77.7% accuracy. Genome-changed analysis shows methylation tools detect more chromosomal territory than SNP in this case (ratios >1.2×). Manual verification shows that the modification in the genome of case 9337 is focal, but just that the segmentation of the SNP data seems to be very fine, leading to an inflated GI. This however does show that the structure of the CNVs is also important to accuracy as much as the algorithm is.

¶6 — Gene discovery implications: The methylation platform reliably detects large canonical events (CDKN2A, TP53) but fails on focal events (MYOCD, CCNE1, MYC). MYOCD concordance of 36% is particularly concerning given its biological relevance in LMS.

¶7 — Recommendations

Cohort	Recommendation
LM (benign)	All three tools clinically adequate. MethylMaster/Sesame preferred. Can be used without SNP validation for most research applications.
LMS (malignant)	Should not replace SNP arrays for individual case-level decisions. Appropriate for large-scale population-level analyses.
Tool selection	No universal winner. Conumee edges out in LMS; MethylMaster/Sesame lead in LM.
GI threshold	Cases with high GI might be high-risk for low methylation-platform accuracy.

¶8 — Limitations

- Small cohort: 13 LMS and 9 LM cases. Case-level conclusions (9337, 9202, 9203) are hypothesis-generating.
- Single platform: all methylation data from EPIC array. Used for normalization and to prevent bias from different platforms, performance may differ on 450k (might be contraindicated by the Smith paper) or next-generation arrays.
- No CBS alternatives tested: SLM, GISTIC, or HMM-based segmentation are future directions.

5. Conclusion

1. **Finding:** Methylation-based CNV calling achieves high accuracy in benign LM tumours (92–96%) but performs near chance in malignant LMS (53–58%).
 2. **Mechanism:** This gap is explained by the Genomic Index, accuracy is inversely correlated with genomic complexity ($r = -0.66$), and is not sensitive to bin size changes.
 3. **Gene level:** Large canonical events (CDKN2A, TP53) are reliably detected, while focal events (MYOCD, CCNE1, MYC) are not.
 4. **Implication:** These findings define the conditions under which methylation-based CNV calling is accurate and posit that GI is a major condition in its performance.
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6. Figure List

Figures already produced and in poster

Figure	Description	Where to use
Fig. 1	Genomic Index method diagram	Methods
Fig. 2	Gene concordance method diagram	Methods
Fig. 3	Whole-dataset accuracy method diagram	Methods
Fig. 4	Gene concordance heatmap — 7 genes $\times$ 12 method/bin combos	Results §3.4
Fig. 5	Bin-wise accuracy bar chart — LMS cases	Results §3.1, §3.2
Fig. 6	Bin-wise accuracy bar chart — LM cases	Results §3.1
Fig. 7	LMS and LM metrics heatmap — clustered by GI and performance	Results §3.3

Unused Figures (Not in Poster)

Figure	Data needed	Key message
Per-case segment plots comparing tools and SNP	GenomicIndexLMS.csv	compression of amplitude by all tools
Side-by-side: LMS vs LM mean accuracy by tool	accuraciesLMS.csv + accuraciesLM.csv	
Genome Changed Bar Graph	genomeChangedLMS.csv + genomeChangedLM.csv	Null result — flat trajectories

Potential figures to create and add

Figure	Data needed	Key message
Scatter: SNP GI vs mean accuracy (LMS cases, annotated)	GenomicIndexLMS.csv + accuraciesLMS.csv	$r = -0.659$; annotate 9337, 9202/9203, 9338/9328/9350
Genome-changed violin: LMS vs LM per tool	genomeChangedLMS.csv + genomeChangedLM.csv	MethylMaster 96% territory vs Sesame 49%
Accuracy line chart: bins 10 kb→1 Mb per tool (LMS)	accuraciesLMS.csv	Null result — flat trajectories

7. Statistics Reference

Accuracy

Metric	Value	Source
LMS mean accuracy — Conumee	58.4%	accuraciesLMS.csv
LMS mean accuracy — MethylMaster	54.8%	accuraciesLMS.csv
LMS mean accuracy — Sesame	52.8%	accuraciesLMS.csv
LM mean accuracy — MethylMaster	96.4%	accuraciesLM.csv
LM mean accuracy — Sesame	95.9%	accuraciesLM.csv
LM mean accuracy — Conumee	91.6%	accuraciesLM.csv
LM median accuracy (all tools)	>99.8%	accuraciesMedianLM.csv
LMS accuracy range by bin (all tools)	54.2–56.7%	accuraciesLMS.csv

Genomic Index

Metric	Value	Source
Mean SNP GI — LMS cohort	~1,272 (default bin)	GenomicIndexLMS.csv
Mean SNP GI — LM cohort	~14 (default bin)	LMGenomicIndex.csv
Highest SNP GI — case 9350	3,463	GenomicIndexLMS.csv
GI compression — MethylMaster	21.6% of SNP GI	GenomicIndexLMS.csv
GI compression — Conumee	16.2% of SNP GI	GenomicIndexLMS.csv
GI compression — Sesame	12.8% of SNP GI	GenomicIndexLMS.csv
GI-accuracy Pearson r (LMS)	-0.659	Derived

Metric	Value	Source
GI-accuracy Pearson r (LM)	-0.652	Derived

Genome Changed

Metric	Value	Source
Mean genome changed — SNP (LMS)	~34% across cases	genomeChangedLMS.csv
MethylMaster detection rate	96.4% of SNP territory	genomeChangedLMS.csv
Conumee detection rate	76.1% of SNP territory	genomeChangedLMS.csv
Sesame detection rate	49.2% of SNP territory	genomeChangedLMS.csv
9337 MethylMaster ratio	1.24× (detects more than SNP)	genomeChangedLMS.csv
9337 Conumee ratio	1.26× (detects more than SNP)	genomeChangedLMS.csv

Gene Concordance

Gene	Concordance	SNP status (LMS)	Clinical relevance
CDKN2A	87%	Deleted 12/13	Tumour suppressor
TP53	85%	Amplified 9/13	Tumour suppressor
PTEN	74%	Mixed	Tumour suppressor
RB1	62%	Mixed	Cell cycle regulation
MYC	41%	Focal amplification	Oncogene
CCNE1	41%	Focal amplification	Cell cycle
MYOCD	36%	Focal	LMS-specific

All data can be found in FinalSet for reverification.